

STATISTICS + PROBABILITY + SQL

1. Statistics -

The science of collecting, organizing, analyzing, interpreting and presenting data.

2. Descriptive statistics -

check 22 from previous day.

3. Inferential statistics -

Methods used to make ^{or} conclusions / predictions about a population using a sample.

4. Population - The complete set of individuals / observations being studied.

5. Sample -

check 56.

6. Parameter -

A numerical value describing a characteristic of an entire population.

example: population mean = μ

7. Statistic -

A numerical value calculated from a sample.

example: sample mean = $\bar{x}$

8. Variable -

A measurable characteristic that can take different values.

9. Independent variable - The input/exploratory variable believed to influence another variable.
10. Dependent variable - The output/response variable whose value depends on other variables.
11. Qualitative variable - A variable representing labels.
12. Quantitative variable - A variable represented using measurable numerical values.
13. Discrete variable - A numerical variable containing countable values.

examples no. of students

14. Continuous variable - A numerical var that can take any value within a range.

examples: weight, height, temperature.

CENTRAL TENDENCY

15. Mean -

check 23.

16. Weighted mean -

An avg where each value contributes according to an assigned weight.

$$\bar{x} = \frac{\sum W_n x_n}{\sum W_n}$$

multiply each value by its weight and add all. $\rightarrow$ $\sum W_n x_n$
 add all the individual weights $\leftarrow$ $\sum W_n$

17. Median - check 24.

18. Mode - check 25.

19. Geometric mean -

The n th root of the product of ' n ' positive values and is useful for growth rates.

$$GM = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n} = (x_1 \times x_2 \times \dots \times x_n)^{1/n}$$

example -

$$4, 9 \rightarrow 2 = n$$

$$\sqrt[2]{4 \times 9} = \sqrt[2]{36} = 6$$

20. Harmonic mean -

An avg useful for rates and ratios

$$HM = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum \left(\frac{1}{x_i} \right)}$$

$$\text{for two numbers: } \frac{2ab}{a+b} = \frac{(GM)^2}{AM} = \frac{(\sqrt{ab})^2}{\left(\frac{a+b}{2}\right)}$$

DISPERSION

21. Range - check 26.

22. Variance - check 27.

23. Sample variance - variance estimated from a sample using $(n-1)$ in the denominator.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

sample mean

24. Standard deviation - check 28.

25. IQR - check 20.

26. Coefficient of variation -

measures variability relative to the mean.

$$cv = \left(\frac{\text{std}}{\text{mean}} \right) \times 100\%$$

POSITIONAL MEASURES

27. Percentile - check 29.

28. Quartile - check 30.

29. Decile - A value dividing sorted data into 10 equal sections.

30. Z-score -

measures how many standard deviations a value lies above or below the mean.

$$Z = \frac{x - \mu}{\sigma}$$

• $z = 0 \rightarrow$ at mean

• $z = 1 \rightarrow$ one SD above mean

• $z = -2 \rightarrow$ two SD below mean

DISTRIBUTIONS

31. Probability distribution -

describes how probabilities are assigned across possible values of a random variable.

32. Discrete probability .d -

A probability distribution for countable outcomes

33. Continuous p.d -

for values that can vary continuously.

34. Normal distribution - check 32.

35. Standard normal .d -

with mean 0 and std 1.

36. Empirical Rule -

for approx. normal data :

68% $\longrightarrow$ within 1 SD.

95% $\longrightarrow$ " 2 SD.

99.7% $\longrightarrow$ " 3 SD.

37. Uniform distribution - where all outcomes in the allowed range have equal probability density.

38. Bernoulli distribution - models one experiment with exactly 2 outcomes.

example : success/failure , $\overset{\curvearrowright}{0/1}$

39. Binomial distribution - models the no. of successes in a fixed no. of independent Bernoulli trials.

40. Poisson .d - models how many times an event occurs within a fixed interval.

- examples -
- calls per hour
 - website errors per day
 - accidents per month

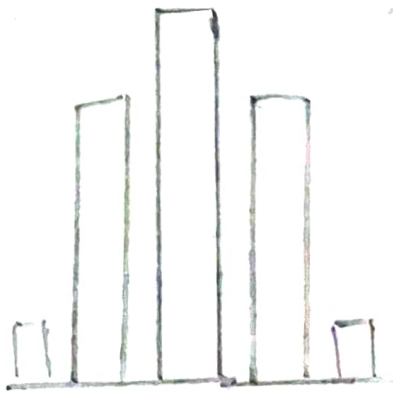
41. Exponential distribution -

Models the waiting time btw independent Poisson events

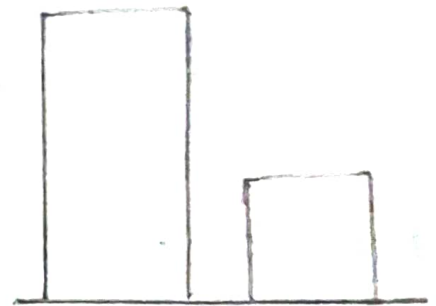
42. Skewness - check 33:

43. Kurtosis - Measures the heaviness of a distribution's tails relative to a normal distribution.

uniform distribution

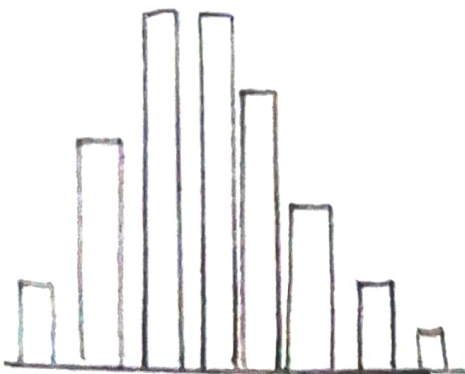


binomial dis -

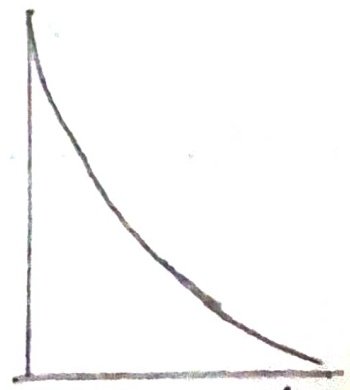
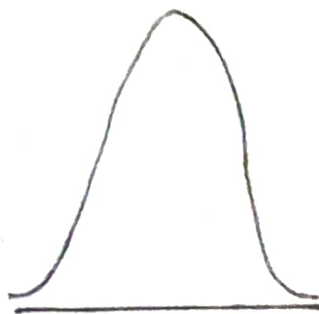


bernoulli distribution

Poisson distribution



normal distribution



exponential distribution

PROBABILITY

44. Probability - A numerical measure of how likely an event is to occur.

$$= \frac{n(A)}{n(S)} = \frac{\text{favourable outcomes}}{\text{total outcomes}}$$

$$P(A) = 1 - P(A')$$

45. Experiment - A process that produces one or more uncertain outcomes.

example: tossing a coin, rolling a die

46. Outcome - A possible result of an experiment.

47. Sample space - The set of all possible outcomes of an experiment.

example: coin toss - $S = \{H, T\}$

48. Event - A subset of the sample space containing outcomes of interest.

49. Complement - The event that A does not occur

$$P(A') = 1 - P(A)$$

50. Mutually Exclusive events - Events that cannot occur at the same time.

$$P(A \text{ or } B) = P(A) + P(B)$$

$$P(A \text{ and } B) = 0$$

51. Independent events - Events where the occurrence of one does not affect the probability of the other.

$$P(A \cap B) = P(A) \cdot P(B)$$

52. Dependent events - Events where the occurrence of one changes the probability of another.

53. Addition Rule - Calculates the probability that A/B occurs

$$P(A \cup B) = P(A) + P(B) - P(A \text{ and } B)$$

54. Multiplication Rule - Calculates the probability that both A and B occur.

$$P(A \text{ and } B) = P(A) \times P(B/A) - \text{dependent}$$

$$P(A \cap B) = P(A)P(B) - \text{independent}$$

55. Conditional probability - The probability of event A occurring given that event B has already occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

56. Bayes' Theorem -

Updates the probability of a hypothesis after observing new evidence.

$$P(A|B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$\text{Posterior} = \frac{\text{likelihood} \times \text{Prior}}{\text{evidence}}$$

57. Prior probability - The probability of a hypothesis before observing new evidence.
58. Likelihood - The prob of observing the evidence assuming a particular hypothesis is true.
59. Posterior probability - The updated prob of a hypothesis after considering evidence.
60. Random variable - A variable that assigns a numerical value to each possible outcome of a random experiment.
61. Expected value - The long-run avg value of a random variable.

$$E(x) = \sum x \cdot P(x)$$

↑ each possible outcome
↓ probability of outcome x occurs.

62. Covariance - check 35.
63. Correlation - check 34.
64. Pearson correlation - Measures linear correlation btw 2 continuous variables.

65. Spearman Correlation - Measures monotonic relationships using ranked values instead of raw values.

66. Correlation ∇ Causation
 $\downarrow$ $\downarrow$
 association means one variable directly influences another.

SAMPLING

67. Sampling - check 54.

68. Simple random sampling

69. Systematic sampling 71. Cluster sampling

70. Stratified sampling 72. Convenience sampling

43. Snowball sampling

74. Sampling Bias - check 58.

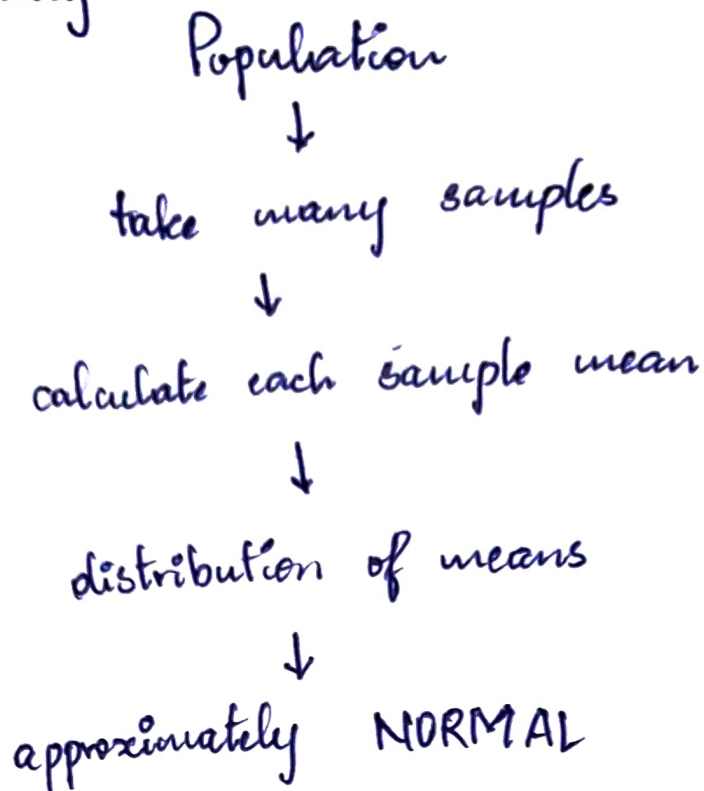
45. Sampling error -

The diff b/w a sample statistic and the true population parameter caused by random sampling.

CENTRAL LIMIT THEOREM & ESTIMATION

46. Central limit Theorem - States that the sampling distribution of sample mean becomes approx. normal as sample size $\uparrow$, even when population is not normal.

Conceptually -



77. Standard Error -

Measures how much a sample statistic is expected to vary from sample to sample.

$$\text{SEM} = \frac{\sigma}{\sqrt{n}} \quad \bigg| \quad \frac{s}{\sqrt{n}} \quad \text{std}$$

↓
SE of mean

78. Point estimate -

A single numerical value used to estimate an unknown population parameter.

example : sample mean estimating population mean.

79. Interval Estimate -

A range of values likely to contain an unknown population parameter.

80. Confidence interval -

A range constructed from sample data that estimates a population parameter with a chosen confidence level.

basic form:

point estimate $\pm$ Margin of error

81. Confidence level -

The long run proportion of similarly constructed intervals expected to contain the true parameter.

common levels - 90%, 95%, 99%.

82. Margin of error -

The amount added to and subtracted from a point estimate to form a confidence interval.

HYPOTHESIS TESTING -

83. Hypothesis testing - A statistical procedure used to evaluate whether sample evidence supports a claim about a population.

84. Null hypothesis (H₀) - The default claim, usually stating that there is no effect, difference or relationship.

85. Alternative Hypothesis (H_1) - The competing claim stating that an effect, diff- / relationship exists.

86. Significance level (α) - The max acceptable probn of rejecting a true null hypothesis.

$$\alpha = 0.05$$

87. P-value - The probability of observing results at least as extreme as the sample result if the null hypothesis were true.

$$P < \alpha \rightarrow \text{reject } H_0$$

$$P \geq \alpha \rightarrow \text{fail to reject } H_0$$

** A p-value is not the probn that H_0 is true.

88. Type I Error (α) - Rejecting H_0 when it actually is true.

$\rightarrow$ False +ve.

89. Type II Error (β) - Failing to reject the H_0 when the alternative is actually true.

$\rightarrow$ False -ve

90. Statistical power - The probn of correctly detecting a real effect.

$$\text{Power} = 1 - \beta$$

91. One-tailed Test -

Tests whether a parameter differs in one specified direction.

eg: mean > 50

92. Two-tailed test -

Tests whether a parameter differs in either direction.

eg: Mean $\neq 50$

COMMON STATISTICAL TESTS

93. Z-test : $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

tests differences involving means when population variance is known or sample size is sufficiently large under appropriate assumptions.

94. T-test :

tests differences b/w means when population variance is unknown (smaller samples)

95. One-sample t-test: compares the mean of a single group to a known value/reference mean

96. Independent t-test: compares means of 2 independent groups.

example $\rightarrow$ Avg Salary: Team A vs Team B

97. Paired t-test : compares 2 related measurements from the same subjects.

example → before training Vs after training

98. Chi-Square test :

used to check if your observed data matches expected values.

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

99. ANOVA :

tests whether there are statistically significant mean differences among three / more groups.

100. F-test : compares variances / forms the basis of tests such as ANOVA.

$$F = \frac{S_1^2}{S_2^2} \quad , \quad S_1, S_2 \text{ are sample variances.}$$

SQL

101. SQL - Structured Query language used to store, retrieve, manipulate, and manage relational db data.

102. Database - An organized collection of data stored electronically.

103. Relational database - organizes data into tables

connected through relationships.

104. Table - A collection of related records stored as rows and columns.
105. Row - A single record in a db table.
106. Column - A field representing one attribute of the records in a table.
107. Primary key - A column/key set of columns that uniquely identifies each row in a table.

Properties: Unique

NOT NULL

108. Foreign key - A column that references a primary or unique key in another table to create a relationship.
109. SQL query - A command used to retrieve or manipulate database data.

SQL COMMAND GROUPS

110. DDL -

Data Definition language used to define/modify database structures.

• CREATE

• DROP

• ALTER

• TRUNCATE

111. DML -

Data Manipulation language used to modify table data.

- INSERT
- UPDATE
- DELETE

112. DQL -

Data Query language primarily to retrieve data.

- SELECT

113. DCL -

Data Control language used to manage db permissions.

- GRANT
- REVOKE

114. TCL -

Transaction Control language used to manage transactions.

- COMMIT
- ROLLBACK
- SAVEPOINT

BASIC SQL

115. SELECT -

Retrieves one / more columns from a table

SELECT name, salary

FROM employees;

116. SELECT * - retrieves all columns.

```
SELECT *  
FROM employees;
```

117. DISTINCT - removes duplicate values

```
SELECT DISTINCT department  
FROM employees;
```

118. WHERE - filters rows according to a condition.

```
SELECT *  
FROM employees  
WHERE salary > 50000;
```

119. Comparison operators - compare SQL values.

=
!=
<>
>
<
>=
<=

120. AND - requires all specified conditions to be true.

```
WHERE age > 25  
AND salary > 50000;
```

121. OR - requires at least one specified condition to be true

```
WHERE city = 'Delhi'  
OR city = 'Rajpur';
```

122. NOT - Negates a condition

WHERE NOT department = 'HR';

123. BETWEEN - filters values within an inclusive range.

WHERE salary BETWEEN 40000 AND 80000;

124. IN - tests whether a value matches any value in a specified list.

WHERE city IN ('Delhi', 'Pune', 'Hyd');

125. LIKE - searches text using pattern matching.

WHERE name LIKE 'A%';

126. Wildcard % -

% : represents zero/more characters.

'A%' → starts with A

'%a' → ends with a

'%an%' → contains an

127. Wildcard - -

- : represents exactly 1 character

WHERE name like 'A_';

128. IS NULL - rows that have missing values.

WHERE salary IS NULL;

129. IS NOT NULL - finds rows where value is present.
WHERE salary IS NOT NULL

SORTING & LIMITING

130. ORDER BY - sorts query results by one/more columns.

```
SELECT *  
FROM employees  
ORDER BY salary;
```

131. ASC - sorts values in ascending order.

```
ORDER BY salary ASC;
```

132. DESC - sorts values in descending order.

```
ORDER BY salary DESC;
```

133. LIMIT - restricts no. of rows returned by a query in databases that support it.

```
SELECT *  
FROM employees  
LIMIT 5;
```

AGGREGATE FUNCTIONS

134. Aggregate function - performs a calculation over multiple rows and returns a summarized result.

135. COUNT - returns no of rows or non-null values.

```
SELECT COUNT(*)  
FROM employees;
```

136. SUM - calculates the total of a numerical column.

```
SELECT SUM(salary)
FROM employees;
```

137. AVG - calculates the average of column

```
SELECT AVG(salary)
FROM employees;
```

138. MIN - returns the smallest value.

```
SELECT MIN(salary)
```

139. MAX - returns the largest value.

```
SELECT MAX(salary)
```

GROUPING

140. GROUP BY - groups rows sharing the same values
so aggregate calculations can be performed per group.

```
SELECT department, AVG(salary)
FROM employees
GROUP BY department;
```

141. HAVING - filters grouped results after aggregation.

```
SELECT department, AVG(salary)
FROM employees
GROUP BY department
HAVING AVG(salary) > 60000;
```

**
memory:

WHERE → filters rows before grouping

HAVING → filters groups after grouping

JOINS

142. Join - combines rows from multiple tables using related columns.

143. INNER JOIN - matching values

```
SELECT *  
FROM emp e  
INNER JOIN department d  
ON e.department_id = d.department_id;
```

144. LEFT JOIN - all rows from left table and matching rows from right.

145. RIGHT JOIN - all rows from right table and matching rows from left table.

146. FULL OUTER JOIN - returns all rows from both tables.

147. CROSS JOIN - return cartesian product of 2 tables.

TABLE A = 3 rows

TABLE B = 4 rows

then :

CJ = 12 rows

148. SELF JOIN - joins a table with itself

useful for :

- employees - manager relationship

- hierarchies

SUBQUERIES

149. Subquery - A query nested inside another.

```
SELECT * FROM employees  
WHERE salary >  
(  
    SELECT AVG(salary)  
    FROM employees  
);
```

150. Correlated subquery -

A subquery that depends on values from the outer query and may execute logically once per outer row.

CTE

151. Common Table Expression - A temporary named result set defined using WITH that improves query organization and readability.

```
WITH high-salary AS (  
    SELECT * FROM employees  
    WHERE salary > 70000  
)  
SELECT * FROM high-salary;
```

152. Recursive CTE - A CTE that repeatedly references itself and is useful for hierarchical or recursive data.

CONDITIONAL SQL

153. CASE - adds conditional logic inside SQL queries.

```
SELECT name,  
       CASE  
         WHEN salary >= 80000 THEN 'High'  
         "      " 50000      " 'Medium'  
         ELSE 'Low'  
       END AS salary-level  
FROM employees;
```

SET OPERATIONS

- 154. UNION - combines results from multiple queries and removes duplicates.
- 155. UNION ALL - combines query results while keeping duplicates.
- 156. INTERSECT - returns rows common to both query results in databases that supports it.
- 157. EXCEPT - returns rows from the 1st query that are absent from the second query.

WINDOW FUNCTIONS

- 158. Window function - performs calculations across related rows without collapsing them into one row per group.

159. OVER - defines the set of rows used by a window function.

AVG(salary) OVER()

160. PARTITION BY -

divides rows into groups for a window function while preserving individual rows.

AVG(salary) OVER(

PARTITION BY department

)

161. ROW_NUMBER - assigns a unique sequential number to rows within an ordered partition.

ROW_NUMBER() OVER(

ORDER BY salary DESC

)

162. RANK - assigns ranking values and leaves gaps after ties.

example :
1
2
2
4

163. DENSE_RANK - assigns rankings without gaps after ties.

example :
1
2
2
3

164. LAG - accesses a value from previous row within a window.

- useful for: previous month's sales, transactions.
166. LEAD - accesses a value from following row within a window.

DATA MODIFICATION

166. INSERT - adds new rows.

```
INSERT INTO employees (name, salary)
VALUES ('Asha', 60000);
```

167. UPDATE - modifies existing rows.

```
UPDATE emp
SET salary = 65000
WHERE employee-id = 1;
```

188. DELETE Vs TRUNCATE Vs DROP

→ table remains	→ table remains	→ removes table
→ removes selected or all rows	→ removes all rows	itself

```
DELETE FROM employees
WHERE employee-id = 1;
```

TABLE CREATION

189. CREATE TABLE - creates a new db table.

```
CREATE TABLE employees (
    employee-id INT,
    name VARCHAR(100),
    salary DECIMAL(10,2)
);
```

170. ALTER TABLE - modifies the structure of existing table.

ALTER TABLE employees

ADD email VARCHAR(100);

CONSTRAINTS

171. Constraint - A rule applied to table columns to enforce valid and consistent data.
172. NOT NULL - prevents a column from storing NULL values.
173. UNIQUE - prevents duplicate values in set of columns.
174. PRIMARY KEY -
Uniquely identifies each row and does not allow NULL values.
175. FOREIGN KEY - enforces a relationship b/w tables.
176. CHECK - restricts values according to a specified condition.
CHECK (salary > 0)
177. DEFAULT - supplies a predefined value when no explicit value is provided.

TRANSACTIONS

178. Transaction - A logical unit of one or more db. operations executed together.
179. COMMIT - permanently saves changes made during a transaction.
180. ROLLBACK - reverses uncommitted transaction changes.

181. SAVEPOINT - creates a point inside a transaction to which changes can later be rolled back.

ACID

182. ACID Properties - A set of properties that ensure reliable db transactions.

1. $A \rightarrow$ Atomicity - A transaction is completed entirely or not executed at all.
2. $C \rightarrow$ Consistency - A transaction moves the db from one valid state to another.
3. $I \rightarrow$ Isolation - Concurrent transactions behave as though they were executed independently.
4. $D \rightarrow$ Durability - Once a transaction is committed, its changes remain permanent even after failures.

NORMALIZATION

183. Database Normalization -

Organizing relational tables to reduce redundancy and prevent update anomalies.

184. First Normal Form — 1NF

1NF : Requires atomic values and no repeating groups in a table.

185. Second Normal Form — 2NF

Requires 1NF and removal of partial dependency on part of a composite key.

186. Third Normal Form — 3NF

Requires 2NF and removal of transitive dependency between non-key attributes.

187. BCNF - A stronger version of 3NF where every determinant must be a candidate key.

188. Denormalization - Intentionally introduces redundancy to improve read performance in some db designs.

INDEXING

189. Index —

A database structure that speeds up data retrieval at the cost of additional storage and slower writes.

190. Clustered Index - Determines the physical/primary storage order of table data in db systems that support it.

191. Non-Clustered Index - Stores a separate search structure containing keys and references to table rows.